Literature Review Notes

Paper notes

Separate reviews are stored in `Paper_Notes/`, with an index in its `README.md`. Each answers the three per-paper questions; the synthesis below draws on those reference papers. Plain-English definitions are collected in the medical and neuroscience nomenclature.

General questions

What can or is measured by optogenetics or deep brain stimulation?

Neural activity is measured, and firing rates can be estimated from those recordings. In the optogenetic experiments of Zaaimi et al., multicontact electrodes record both LFPs and single-neuron spikes; Fig. 3 reports spike-rate responses during stimulation [10, PDF pp. 2, 5]. Electrical-stimulation experiments used by Nevado Holgado et al. likewise record STN/GPe firing rates, including GPe responses to STN stimulation [15, PDF pp. 3–4]. The DBS review by Neumann et al. summarizes measured single-neuron firing responses at several stimulation targets [12, PDF pp. 4–6].

For a population represented by n_i recorded neurons, a possible measurement construction is the average spike count per unit time:

$$\widehat{r}_i(t) = \frac{1}{n_i \Delta} \sum_{\ell=1}^{n_i} N_{i\ell}((t-\Delta, t]),$$

where $N_{i\ell}$ counts spikes and Δ is the observation window. This estimates a population mean when the recorded neurons are representative. In deviation coordinates it motivates $y = C_y x + \nu$, with C_y selecting or mixing recorded populations and ν representing sampling and measurement error; windowing adds observation dynamics.

LFPs also provide a route to observing rate dynamics. Zaaimi et al. model LFP as a high-pass-filtered combination of excitatory and inhibitory population activity [10, PDF pp. 9, 15]. Hall et al., cited by that paper, demonstrate real-time estimation of individual cortical firing rates from multichannel LFPs using a trained decoder [3]. Optical indicators offer another route to activity estimation [1, 2]. Thus both spike-based rate estimates and calibrated indirect observations can support a firing-rate state model. The observation map depends on the recording method; an unprocessed LFP is not itself a firing rate.

Is state feedback sufficient, or is an observer needed?

State feedback is an explicit measurement assumption. It is experimentally plausible when the states used by the controller are population rates estimated from representative, separately identified recordings of those populations. Sampling error, rate filtering and latency must then be included in the implementation model. Recording neural activity alone does not establish access to every model state.

Evidence for estimation. Zaaimi et al. feed back one filtered LFP channel, while their model contains both E and I [10, PDF pp. 2, 9, 14–15]. Their experiment demonstrates output feedback, rather than separate measurement of both states. Hall et al. demonstrate an LFP-to-firing-rate decoder [3]. More directly, Bolus et al. synthesize linear-quadratic state-feedback gains and

implement optogenetic firing-rate regulation in awake mice using a parameter-adaptive Kalman filter for online state estimation [4, Secs. 2.5–2.6]. This is experimental precedent for combining state-feedback synthesis with an observer; their identified linear model differs from the nonlinear delayed model here.

Implication for this project (derived). For instantaneous measurements $y = C_y x$, a law $u = Kx$ can be implemented directly if its required feedback is reconstructible from y; for example, if $K = LC_y$, then $u = Ly$. Otherwise an observer/decoder must estimate the required state components, provided they can be inferred from the available measurements and dynamics. A mixed LFP can contain enough temporal information for estimation, but this requires an observability or detectability analysis. With an estimate,

$$u = K\widehat{x} = Kx + Ke, \qquad e = \widehat{x} - x.$$

The estimation error enters the closed loop, so the existing state-feedback certificate alone does not certify the observer-based nonlinear delayed system. For PIE feedback using delay histories, those histories are also required; measured rates can be buffered to supply them after initialization.

What can or is controlled by optogenetics or deep brain stimulation?

Stimulation supplies an input to neuronal populations and changes their firing dynamics. Optogenetics targets opsin-expressing neurons or projections. Zaaimi et al. explicitly represent optical drive $S(t)$ as an input to the excitatory population and measure the resulting spiking and LFP activity [10, PDF pp. 2, 5, 15]. The reviewed optogenetic perturbations of STN/GPe also produce recorded changes in neuronal firing rates [14, PDF pp. 6–8].

For electrical stimulation, Nevado Holgado et al. model STN electrode current through an additive input and fit its effect to GPe firing-rate responses [15, PDF pp. 3–4, Eq. 5]. DBS recruits neurons and axons, thereby influencing the stimulated and connected populations [12]. These examples support both input locations in the proposed model: $c^u u$ changes the drive entering a population's nonlinear activation, whereas $b^u u$ represents an additive effective rate drive. Targeting determines which channels are actuated; kinetics and pathway recruitment determine their gains, signs and dynamics.

What causes the neurological symptoms?

For Parkinson's disease, dopamine loss changes population firing rates, synaptic interactions and the temporal organization of basal-ganglia activity. A dynamical firing-rate model can represent both changes in mean activity and collective oscillations through time-varying population rates [15, 16]. Exaggerated beta synchronization is associated particularly with bradykinesia and rigidity; its contribution to symptoms must be distinguished from the mechanism generating it. Motor deficits can precede exaggerated beta [14, 9].

The mechanism is not settled by a single loop: delayed STN–GPe models and models emphasizing pallidostriatal interactions provide different accounts of synchronization [15, 16, 17]. For freezing of gait, cortical PAC is associated with severity, yet DBS improves freezing even at comparable PAC levels. Thus PAC reduction alone does not explain the treatment effect [5].

What is the control objective?

At the neural level, the objective can be to regulate population activity towards a desired rate or trajectory, or to attenuate unwanted oscillatory/bursting dynamics. Grosenick et al.

discuss target neural activity patterns, and Zaaimi et al. demonstrate feedback enhancement and suppression of oscillatory activity [1, 10]. For regulation about a chosen reference rate, a possible design choice is

$$x = r - r^{\mathrm{ref}}, \qquad z = \begin{bmatrix} W_r x \\ W_u u \end{bmatrix}, \qquad C_z = \begin{bmatrix} W_r \\ 0 \end{bmatrix}, \quad D_{zu} = \begin{bmatrix} 0 \\ W_u \end{bmatrix}.$$

Here the weights trade rate deviations against stimulation effort; the reference and weights are design choices. For clinical use, the chosen activity target should be linked to an outcome such as improved movement initiation or reduced freezing [6, 5].

Can the system be modeled as follows? If so give the parameterization

Yes: key references explicitly use population firing-rate models with stimulation inputs and neural observations. Zaaimi et al. use a two-population Wilson–Cowan model for their optogenetic experiments; Nevado Holgado et al. use delayed STN–GPe rate dynamics constrained by recordings. Azizpour Lindi et al. also use a rate model alongside a spiking network [10, 15, 16]. This supports the proposed model class as a population-level description; the populations and coefficients depend on the preparation.

Let $x(t) \in \mathbb{R}^{N_x}$, $w(t) \in \mathbb{R}^{N_w}$, and $u(t) \in \mathbb{R}^{N_u}$ denote the state, disturbance, and control. Let $v_i(t)$ be the input to f_i, $z(t) \in \mathbb{R}^{N_z}$ the performance output, and $y(t) \in \mathbb{R}^{N_y}$ the measurement:

$$\begin{aligned}
\dot{x}_i(t) &= a_i x_i(t) + f_i(v_i(t)) + \sum_{j=1}^{N_x} b_{ij}^x x_j(t-\hat{\tau}_{ij}) + \sum_{j=1}^{N_w} b_{ij}^w w_j(t-\overline{\tau}_{ij}) + \sum_{j=1}^{N_u} b_{ij}^u u_j(t-\tilde{\tau}_{ij}),\\
v_i(t) &= \sum_{j=1}^{N_x} c_{ij}^x x_j(t-\hat{\tau}_{ij}) + \sum_{j=1}^{N_w} c_{ij}^w w_j(t-\overline{\tau}_{ij}) + \sum_{j=1}^{N_u} c_{ij}^u u_j(t-\tilde{\tau}_{ij}),\\
z(t) &= C_z x(t) + D_{zu} u(t),\\
y_i(t) &= \sum_{j=1}^{N_x} c_{ij}^y x_j(t-\tau_{ij}^y) + \sum_{j=1}^{N_w} d_{ij}^{yw} w_j(t-\tau_{ij}^{yw}).
\end{aligned}$$

The first two equations use $i = 1, \ldots, N_x$; the measurement equation uses $i = 1, \ldots, N_y$. Local state measurements are obtained by choosing the corresponding c_{ij}^y nonzero. All delays are constant and nonnegative, with the required state and input histories prescribed.

Nonlinearity assumptions. In equilibrium-deviation coordinates, assume f_i is memoryless, time-invariant, and satisfies $f_i(0) = 0$. It is monotone nondecreasing and slope-restricted on an interval $\mathcal{I}_i^{\mathrm{sl}}$ containing zero:

$$0 \le \frac{f_i(v) - f_i(q)}{v - q} \le \ell_i, \qquad v, q \in \mathcal{I}_i^{\mathrm{sl}}, \quad v \ne q, \quad 0 \le \ell_i < \infty.$$

On a separate interval $\mathcal{I}_i^{\mathrm{sec}}$ containing zero, assume the sector bound

$$0 \le \frac{f_i(v)}{v} \le \alpha_i, \qquad v \in \mathcal{I}_i^{\mathrm{sec}} \setminus \{0\}, \quad 0 \le \alpha_i < \infty.$$

Oddness is not required.

Optogenetic example: Zaaimi et al.

Their Wilson–Cowan model has excitatory and inhibitory activities E, I, optical input S, tonic drives P, Q and fluctuating inputs η_E, η_I [10, PDF p. 15, Eqs. 3–4]:

$$\dot{E} = \frac{-E + \sigma(aE - bI + S + P)}{\tau_E} + \eta_E,$$
$$\dot{I} = \frac{-I + \sigma(cE - dI + Q)}{\tau_I} + \eta_I.$$

Around an equilibrium, take $x = (E - E^*, I - I^*)^\top$, $u = S - S^*$, $a_i = -1/\tau_i$, and

$$[c^x_{ij}] = \begin{bmatrix} a & -b \\ c & -d \end{bmatrix}, \qquad c^u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad b^x = b^u = 0, \qquad f_i(v) = \frac{\sigma(q_i^* + v) - \sigma(q_i^*)}{\tau_i}.$$

All intrinsic delays are zero in this example. Additive disturbance channels can represent fluctuating inputs; the paper simulates them stochastically. Its logistic has derivative between zero and $1/4$, giving $\ell_i = \alpha_i = 1/(4\tau_i)$ for the centred nonlinearities. The four letters a, b, c, d above are that paper's coupling parameters; the numerical simulation values are referred to its Supplementary Table 2.

The simulated measurement is a high-pass-filtered combination of E and I, passed through the experimental feedback algorithm [10, PDF pp. 9, 15]. Thus the paper connects population dynamics, an optical input and a neural observation in one closed loop. The filtered measurement can be represented by augmenting the model with observation-filter states. The recorded spikes additionally allow rate estimates for the sampled neurons; the two model population activities are not separately measured as complete populations.

Electrical-stimulation example: Nevado Holgado et al.

Nevado Holgado et al. [15, PDF pp. 2–4, Eqs. 2–5 and Tables 1, 3] provide a concrete two-population example. Let $x = (r_S - r_S^*, r_G - r_G^*)^\top$ denote STN and GPe firing-rate deviations from an equilibrium, and let q_i^* be its total input. Choose

$$a_i = -1/\tau_i, \qquad b^x_{ij} = b^w_{ij} = 0,$$
$$[c^x_{ij}] = \begin{bmatrix} 0 & -w_{GS} \\ w_{SG} & -w_{GG} \end{bmatrix}, \qquad [\hat{\tau}_{ij}] = \begin{bmatrix} 0 & T_{GS} \\ T_{SG} & T_{GG} \end{bmatrix},$$
$$f_i(v) = \frac{F_i(q_i^* + v) - F_i(q_i^*)}{\tau_i}, \quad F_i(q) = \frac{M_i}{1 + \frac{M_i - B_i}{B_i} e^{-4q/M_i}}.$$

The nominal cortical and striatal drives are $(\mathrm{Ctx}, \mathrm{Str}) = (27, 2)$ spikes/s. If their deviations are included as w, take $[c^w_{ij}] = \mathrm{diag}(w_{CS}, -w_{XG})$ with zero input delays. The remaining parameters are

$$(\tau_S, \tau_G, T_{SG}, T_{GS}, T_{GG}) = (6, 14, 6, 6, 4) \text{ ms},$$
$$(M_S, B_S, M_G, B_G) = (300, 17, 400, 75) \text{ spikes/s},$$
$$(w_{SG}, w_{GS}, w_{GG}, w_{CS}, w_{XG}) = \begin{cases} (19, 1.12, 6.60, 2.42, 15.1), & \text{healthy fit}, \\ (20, 10.7, 12.3, 9.2, 139.4), & \text{diseased fit}. \end{cases}$$

Use consistent time units. The sigmoid satisfies $0 \le F_i' \le 1$; therefore the centred f_i satisfies the stated assumptions globally with $\ell_i = \alpha_i = 1/\tau_i$ (derived from the reported sigmoid). For their electrode-pulse model, u is current and enters additively through $b^u = (\mathrm{Stim}/\tau_S, 0)^\top$, with $c^u = 0$ and zero input delay; the healthy-data fit gives $\mathrm{Stim} = 4.6 \times 10^3$ (spikes/s)/mA. This supplies an experimentally fitted example of the additive stimulation channel.

For spike-derived rate measurements, C_y can select or average the recorded populations, with sampling error and observation filtering included as above. If the recorded signal is LFP, its calibrated observation model must be supplied separately. Performance weights can be chosen using the proposed z above. More detailed preparations may require additional populations, opsin states or synaptic filters [1, 16, 17].

How can I optimally parameterize a PIE Zames–Falb multiplier?

Scope of the answer. The neural reference papers do not derive an optimal PIE multiplier. The family below is a derived candidate whose weights can shape its frequency response. Optimality would be relative to a specified robustness/performance objective, polynomial degree and feasible multiplier class.

This derived construction has scalar input and output and distributed states. Use polynomial profiles on $s \in [0,1]$, a finite number n of state components, and a scalar feedthrough decision variable d, with

$$\boxed{d > 0, \qquad a_j(s) > 0, \qquad \beta_j(s) \geq 0 \quad \text{for all } s \in [0,1],\ j = 1, \dots, n.}$$

Continuity on this compact interval gives a uniform lower bound $a_j(s) \geq a_{\min} > 0$ and finite integrals.

PIE realization of M^{-1}

Let $Z = L_2([0,1]; \mathbb{R}^n)$. For $x(t,\cdot) \in Z$ and scalar $u(t), y(t)$, take

$$\dot{x}_j(t,s) = -a_j(s)x_j(t,s) + \beta_j(s)u(t), \qquad x_j(0,s) = 0,$$
$$y(t) = d\,u(t) + \int_0^1 \sum_{j=1}^{n} a_j(s)x_j(t,s)\,ds.$$

Equivalently, $\mathcal{T} = I$ and

$$(\mathcal{A}x)(s) = -\operatorname{diag}(a_1(s), \dots, a_n(s))x(s), \quad (\mathcal{B}u)(s) = \begin{bmatrix}\beta_1(s) & \cdots & \beta_n(s)\end{bmatrix}^\top u,$$
$$\mathcal{C}x = \int_0^1 \sum_j a_j(s)x_j(s)\,ds, \qquad\qquad D = d.$$

With the profiles $a_j(s)$ fixed, $\mathcal{A}, \mathcal{C}$ are fixed, and $\mathcal{B}, D$ are affine in the weight coefficients and d.

Transfer functions and normalization

Writing p for the Laplace variable, define

$$G(p) = \int_0^1 \sum_{j=1}^{n} \beta_j(s)\frac{a_j(s)}{p + a_j(s)}\,ds.$$

Then

$$\boxed{\begin{aligned} M^{-1}(p) &= d + G(p), \qquad & M(p) &= \frac{1}{d + G(p)}, \\ H(p) &= \frac{G(p)}{d + G(p)}, \quad & H^{-1}(p) &= 1 + \frac{d}{G(p)}. \end{aligned}}$$

Use the scaled Zames–Falb normalization $M = d^{-1}(1 - H)$; $d = 1$ recovers $M = 1 - H$. For $G \not\equiv 0$, H^{-1} exists algebraically and may be improper. Spatially varying profiles can give nonrational transfers; $M, M^{-1}, H \in H_\infty$.

Nonnegative impulse response and L_1 condition

Set $b_j(s) = \sqrt{a_j(s)\beta_j(s)}$ and let $\mathcal{A}_a$ be multiplication by $\operatorname{diag}(a_j(s))$ on Z. With $b^*x = \langle b, x\rangle_Z$, define $\mathcal{K} = \mathcal{A}_a + d^{-1}bb^*$. The inversion identity gives

$$H(p) = \frac{1}{d}b^*(pI + \mathcal{K})^{-1}b, \qquad h(t) = \frac{1}{d}\langle b, e^{-\mathcal{K}t}b\rangle_Z \geq 0.$$

The operator $\mathcal{K}$ is self-adjoint and bounded below by $a_{\min} I$, so h decays exponentially. Consequently,

$$\boxed{S := \int_0^1 \sum_j \beta_j(s)\, ds \geq 0, \qquad \|h\|_{L_1} = H(0) = \frac{S}{d+S} < 1.}$$

No upper bound on S, including $S < d$, is required. This is the temporal L_1 norm of a scalar impulse response. Background: Scherer (2022), *Dissipativity and Integral Quadratic Constraints*, Lemma 31 and (S18), p. 132. Its normalized hard IQC assumes a convex-subgradient nonlinearity and zero filter initialization; scaling by $1/d > 0$ preserves the inequality.

Implication for optimization

For fixed positive profiles $a_j(s)$, expand $\beta_j(s)$ in a finite polynomial basis and impose nonnegativity, for example through an SOS certificate on $[0, 1]$. The weight coefficients and d are then candidate decision variables. An SDP requires the *complete* IQC/storage inequality to be affine in all joint decision variables; affinity of $\mathcal{B}, D$ alone does not establish this. Jointly varying $a_j(s)$ and a Lyapunov operator generally introduces products of unknowns. Alternating between fixed-profile and fixed-Lyapunov searches is therefore a possible heuristic when those subproblems are convex, with no guarantee of a global optimum.

Medical and neuroscience nomenclature

Plain-English explanations of the medical and neuroscience terms used in this review, the paper notes and the related reading. Entries are grouped by topic. Frequency bands are approximate: individual papers may use different boundaries.

Term / abbreviation	Meaning

Neurons and cellular activity

Action potential / spike	A brief electrical pulse generated by a neuron and carried along its axon.
Axon	The extension of a neuron that carries spikes towards other cells.
Burst / bursting	A cluster of spikes occurring close together in time.
Cell body / soma	The main part of a neuron, containing its cell nucleus.
Dendrite	A branching extension through which a neuron receives many of its inputs.
Depolarization	A change that makes the voltage inside a neuron less negative relative to its surroundings.
Excitation / excitatory	An effect that makes a neuron more likely to fire. An excitatory neuron usually has this effect on its targets.
Fast-spiking neuron / FSI	A neuron capable of sustained rapid firing. FSI means fast-spiking interneuron, commonly an inhibitory cell.
Firing	A neuron producing action potentials.
Firing rate	The number of spikes per unit time, usually expressed as spikes per second. It is estimated over a time window.
Hyperpolarization	A change that makes the voltage inside a neuron more negative relative to its surroundings.
Inhibition / inhibitory	An effect that makes a neuron less likely to fire; it need not stop firing completely.
Interneuron	A neuron whose connections mainly influence a local circuit. Many cortical interneurons are inhibitory.
Membrane potential	The voltage difference between the inside and outside of a cell.
Neural dynamics	How neural activity changes over time through inputs and interactions within a circuit.
Neural circuit / network	Interconnected neurons or populations that influence one another.
Neural population / ensemble	A group of neurons considered together, for example because they share a location, cell type or activity pattern.
Neuron	A nerve cell that receives, processes and transmits signals.
Neurophysiology	The study of how neurons and the nervous system function.
Parvalbumin neuron / PV neuron	A neuron containing the calcium-binding protein parvalbumin. Cortical PV neurons are commonly fast-spiking inhibitory interneurons.
Population firing rate	The average firing rate of a group of neurons. A recorded sample can estimate it if the sample represents the group.
Pyramidal neuron	A common excitatory cortical neuron named after its roughly pyramid-shaped cell body; "pyramidal" names the cell type, not an oscillation.
Spontaneous / endogenous / native activity	Activity generated by the neural system itself, rather than directly prescribed by an external stimulus.
Spike train	The sequence of times at which a neuron fires.
Tonic / phasic activity	Tonic activity is relatively sustained; phasic activity consists of brief changes or responses.

Synapses, signalling and adaptation

Dopamine	A chemical messenger that changes how neural circuits operate, including circuits involved in movement, motivation and learning.

Term / abbreviation	Meaning (continued)
Dopaminergic	Producing, releasing, responding to or otherwise involving dopamine.
Dopamine depletion / hypodopaminergic	Reduced dopamine availability or signalling in the relevant circuit.
D1 / D2 receptors	Two families of proteins through which dopamine affects cells. Their effects depend on the cell and circuit.
GABA	Gamma-aminobutyric acid: the main inhibitory chemical messenger in the adult brain.
GABAergic	Using GABA to transmit signals.
Glutamate / glutamatergic	Glutamate is the main excitatory chemical messenger in the brain; glutamatergic means using glutamate.
Neurotransmitter	A chemical released by a neuron to communicate with another cell.
Postsynaptic potential	A change in a receiving neuron's membrane voltage caused by synaptic input. It differs from a spike.
Presynaptic / postsynaptic	Presynaptic refers to the sending side of a synapse; postsynaptic refers to the receiving side.
Receptor	A protein that responds to a signalling molecule and changes the receiving cell's activity.
Synapse / synaptic	A connection through which one neuron influences another cell; synaptic means relating to that connection.
Synaptic depression	A decrease in the effect of a synapse during repeated activity. Here "depression" does not mean a mood disorder.
Synaptic plasticity	A change in the strength or effectiveness of a synapse over time.

Brain regions and pathways

Afferent / efferent	Afferent connections arrive at a specified region; efferent connections leave it.
Arkypallidal neuron / Arky	A type of GPe neuron with prominent projections back to the striatum.
Basal ganglia / BG	Interconnected brain structures involved in selecting and regulating actions, as well as other functions.
Cortex / cortical	The brain's outer sheet of neural tissue; cortical means relating to it.
Cortico-subthalamic	Connecting or involving the cortex and the subthalamic nucleus.
Contralateral / ipsilateral	On the opposite side / the same side of the body relative to the structure or intervention being described.
D1-MSN / D2-MSN	Striatal neurons distinguished by dopamine-receptor expression, typically associated with the direct and indirect pathways respectively.
Direct pathway	A basal-ganglia route from the striatum to output nuclei such as GPi/SNr, without passing through GPe.
Dorsal / ventral	In the spinal cord, towards the back / the belly side. Dorsoventral means spanning this direction.
Fibre / tract / projection	An axon, bundle of axons or connection carrying signals between neurons or regions.
GPe	External globus pallidus: a basal-ganglia region interacting with the STN, striatum and other regions.
GPi	Internal globus pallidus: a major basal-ganglia output region and a DBS target. It differs from GPe.
Hippocampus	A brain structure important for memory and spatial navigation.
Hyperdirect pathway	A route carrying signals directly from cortex to STN, bypassing the striatum.
Indirect pathway	A basal-ganglia route involving striatum, GPe and downstream output circuits; the classical route also passes through STN.
Motor cortex / M1	Cortical regions involved in movement; M1 is the primary motor cortex.
MSN / SPN	Medium spiny neuron / spiny projection neuron: a principal inhibitory projection neuron of the striatum.

Term / abbreviation	Meaning (continued)
Nucleus	In brain anatomy, a group of neuron cell bodies. This differs from the nucleus inside an individual cell.
Pallidal / pallidostriatal	Pallidal means relating to the globus pallidus; pallidostriatal refers to connections from pallidum to striatum.
Premotor cortex / SMA	Cortical areas involved in preparing and organizing movements; SMA is the supplementary motor area.
Prototypic neuron / Proto	A major GPe cell type that projects to other basal-ganglia regions, including STN.
Rostral / caudal	Towards the head / towards the tail end of the body or spinal cord.
Sensorimotor	Relating to sensory information and movement control.
SNc	Substantia nigra pars compacta: a region containing dopamine-producing neurons that are particularly affected in Parkinson's disease.
SNr	Substantia nigra pars reticulata: a major basal-ganglia output region, distinct from SNc.
Spinal cord	Neural tissue connecting the brain with the body and containing circuits involved in movement and sensation.
STN	Subthalamic nucleus: a small, mainly excitatory basal-ganglia structure and a common DBS target.
Striatum / striatal	A major input region of the basal ganglia; striatal means relating to it.
Subcortical	Located beneath the cerebral cortex.
Thalamus / thalamocortical	The thalamus is a group of brain nuclei communicating with cortex and other regions; thalamocortical means connecting thalamus and cortex.
Vim	Ventral intermediate nucleus of the thalamus: a stimulation target used in tremor treatment.

Neural signals and rhythms

Alpha / theta activity	Named rhythm bands: alpha is roughly 8–12 Hz; theta is roughly 4–8 Hz, extended to 10 Hz in some reviewed work.
Beta activity	Rhythmic activity at roughly 13–30 Hz; some reviewed papers extend the band to 35 Hz.
Coherence	A frequency-specific measure of how consistently two signals are related. It does not by itself establish causation.
Entrainment	A rhythm becoming aligned with an external rhythm, such as periodic stimulation.
Evoked response	Neural activity occurring in response to a stimulus.
Gamma activity	Rhythmic activity above roughly 30 Hz. Upper limits and the definition of high gamma vary by study.
Granger causality	A test of whether one signal's past improves prediction of another. The name does not imply proof of a biological cause.
LFP	Local field potential: a voltage recorded in neural tissue, reflecting combined nearby electrical currents, especially synaptic activity.
Neural oscillation / rhythm	A recurring rise and fall in neural activity. It can occur in population rates or field potentials.
PAC	Phase-amplitude coupling: a statistical relationship between the phase of a slower rhythm and the amplitude of faster activity.
Phase / amplitude	Phase is position within an oscillation's cycle; amplitude describes the size of its variation.
Readiness potential	A slow electrical signal associated with preparation for voluntary movement.
Spectral / band power	How much signal variation lies within a frequency range. Beta power, for example, measures variation in the beta band.
Synchronization / synchrony	Coordination in the timing of neural activity. Neurons need not have identical average firing rates.

Term / abbreviation	Meaning (continued)
Wilson–Cowan / firing-rate model	A population-level description of how neural activity changes through excitation, inhibition and external inputs; Wilson–Cowan models commonly use coupled excitatory and inhibitory populations.

Recording and imaging methods

Calcium imaging / indicator	Recording activity-related changes in intracellular calcium using a fluorescent sensor; calcium signals are slower than individual spikes.
Connectome / connectivity	A map or description of connections between brain regions. Structural connectivity concerns anatomy; functional connectivity concerns statistical relationships between activity signals.
ECoG	Electrocorticography: recording electrical brain activity using electrodes on the cortical surface.
EEG	Electroencephalography: recording electrical brain activity, usually with electrodes on the scalp.
Electrode / microelectrode	A conducting contact used to record or stimulate electrically; a microelectrode is a small contact suited to localized recordings.
Electrophysiology	Measurement of electrical activity in cells or tissue.
EMG	Electromyography: recording electrical activity associated with muscle activation.
Extracellular recording	Recording from outside cells; depending on the setup, this can capture spikes and field potentials.
Intracranial recording	Recording using electrodes placed inside the skull.
MEG	Magnetoencephalography: measuring magnetic fields produced by brain activity.
MRI / fMRI	Magnetic resonance imaging measures tissue properties; functional MRI uses blood-oxygen-related changes to study brain activity indirectly.
Multi-unit / single-unit activity	Multi-unit activity combines spikes from several neurons; single-unit activity consists of spikes assigned to one neuron.
PET	Positron emission tomography: imaging the distribution of an injected radioactive tracer, for example to study molecular systems.
Polysomnography / PSG	A sleep recording combining signals such as EEG, eye movements, muscle activity and breathing.
sEEG	Stereo-electroencephalography: recording electrical activity with depth electrodes placed within the brain.
Spike sorting	Assigning recorded spikes to putative individual neurons using their waveform and other features.
Tractography	Estimating the courses of nerve-fibre bundles from diffusion MRI data.

Stimulation and optogenetics

Adaptive DBS / aDBS	DBS whose settings change automatically in response to measured signals.
Antidromic / orthodromic	Antidromic spikes travel towards the cell body along an axon; orthodromic spikes travel in its usual output direction.
Archaerhodopsin / Arch / ArchT	Light-driven proton pumps used to inhibit neurons; ArchT is an engineered variant.
Channelrhodopsin / ChR2 / Chronos	Light-sensitive ion channels used to excite neurons; ChR2 and Chronos are examples.
Closed-loop stimulation	Stimulation chosen using ongoing measurements. Its purpose may be to increase, suppress or otherwise study activity.
DBS	Deep brain stimulation: electrical stimulation delivered through implanted electrodes to influence brain circuits.
Halorhodopsin / NpHR / eNpHR	A light-driven chloride pump used to inhibit neurons; eNpHR denotes enhanced versions.
HFS	High-frequency stimulation: a rapid train of stimulation pulses; the exact frequency depends on the experiment.

Term / abbreviation	Meaning (continued)
Neuromodulation	Changing neural activity using an intervention such as electrical stimulation, optical stimulation or a chemical agent.
Open-loop stimulation	Stimulation delivered without adjusting it in response to ongoing measurements of the controlled activity.
Opsin	A light-sensitive protein. In these experiments, introduced opsins allow light to influence neuronal activity.
Optogenetics	Using genetically introduced light-sensitive proteins to manipulate selected cells or their connections with light.
Optrode	A device combining optical stimulation with electrode recording.
Photostimulation	Stimulation using light.
Projection targeting	Selecting neurons or their axons according to which region they connect to.
Pulse width / stimulation amplitude	Pulse width is a pulse's duration; amplitude is its electrical or optical strength.
Stimulation artifact	A recorded signal caused by the stimulation or equipment rather than the neural activity being studied.
Viral vector / opsin expression	A modified virus can deliver genetic instructions; opsin expression means cells produce the instructed light-sensitive protein.

Symptoms, disorders and treatment terms

Akinesia	Difficulty initiating movement, or a marked lack of movement.
Axial symptoms	Problems mainly affecting the trunk, posture, balance or walking.
Biomarker	A measurable biological characteristic used to indicate a condition or its response to an intervention.
Bradykinesia	Slowness of movement, often with progressively smaller or slower repeated movements.
Cognitive	Relating to processes such as attention, memory, learning and decision-making.
Dyskinesia	Involuntary movements; in Parkinson's research, often abnormal movements associated with dopaminergic treatment.
Dystonia	Involuntary muscle contractions causing abnormal postures or repetitive twisting movements.
Epilepsy / seizure	A seizure is an episode of abnormal brain activity; epilepsy is a disorder involving an enduring tendency to have seizures.
Essential tremor	A movement disorder characterized mainly by tremor during posture or action.
Freezing of gait / FoG	A brief inability or marked difficulty to step despite the intention to walk.
Gait	The pattern of walking.
Huntington's disease	An inherited neurodegenerative disorder affecting movement, cognition and behaviour.
Hypokinetic / hyperkinetic	Hypokinetic refers to reduced movement; hyperkinetic refers to excessive or unwanted movement.
Levodopa / L-DOPA	A substance converted into dopamine in the body and used to treat Parkinson's symptoms.
Motor / non-motor symptoms	Motor symptoms affect movement; non-motor symptoms include effects on sleep, mood, thinking and other functions.
Neurodegeneration	Progressive damage to or loss of neurons.
ON / OFF medication	Study conditions describing treatment benefit or medication exposure; the paper specifies which meaning is used. These differ from stimulation ON/OFF.
Parkinson's disease / PD	A progressive neurodegenerative disorder involving dopamine-neuron loss and motor as well as non-motor symptoms.

Term / abbreviation	Meaning (continued)
Parkinsonism / parkinsonian	Parkinsonism is a pattern of movement signs resembling Parkinson's disease; parkinsonian also describes these features in experimental models.
Pathological / physiological	Pathological concerns disease-related dysfunction; physiological concerns normal biological function.
Phenotype	An observable characteristic, such as a movement pattern or disease-related behaviour.
Rigidity	Increased resistance when a joint is moved by someone else; commonly experienced as stiffness.
Tourette syndrome	A condition involving recurrent motor and vocal tics.
Tremor	An involuntary rhythmic shaking movement.
Volitional / voluntary movement	Movement initiated intentionally.

Sleep and experimental terminology

Acute / chronic	Acute usually refers to a short-term condition or experiment; chronic refers to a persistent condition or long-term recording/intervention.
Anaesthetized	Under anaesthesia; the drugs can affect neural activity and responses to stimulation.
Brain slice	A thin piece of brain tissue kept viable outside the animal for experiments.
Biophysical	Describing biological activity through physical mechanisms such as membrane currents and ion channels.
Causal / correlational evidence	Causal evidence supports one process influencing another; correlation describes variables changing together.
Histology	Examining tissue structure, usually with microscopy and stains.
Epidural	Located outside the dura mater, the tough outer membrane surrounding the brain and spinal cord.
In silico / in vitro / in vivo	In silico: in a computer model. In vitro: outside the living organism. In vivo: within a living organism.
NHP	Non-human primate, such as a monkey used in research.
NREM sleep / N1–N3	Non-rapid-eye-movement sleep, subdivided into stages N1, N2 and N3.
REM sleep	Rapid-eye-movement sleep: a sleep state with characteristic brain activity, eye movements and usually reduced muscle tone.
Sleep decoding	Estimating sleep state from recorded signals using an algorithm.
Transgenic animal	An animal whose genetic material has been deliberately altered, for example to express an opsin in selected neurons.
6-OHDA / MPTP model	Experimental Parkinsonian models using substances that damage dopamine systems; their effects do not reproduce every feature of human disease.

Sources. Circuit, recording and optogenetic terminology follows the reviewed papers, particularly [1, 2, 12, 14]. Additional plain-language checks: NINDS neurological glossary, NINDS Parkinson's disease overview, and *Neuroscience*: excitatory and inhibitory postsynaptic potentials.

References

[1] Grosenick et al. (2015). *Closed-Loop and Activity-Guided Optogenetic Control*. Neuron.

[2] Fenno et al. (2011). *The Development and Application of Optogenetics*. Annual Review of Neuroscience.

[3] Hall, Nazarpour and Jackson (2014). *Real-time estimation and biofeedback of single-neuron firing rates using local field potentials*. Nature Communications. Further source cited by Zaaimi et al., ref. 13.

[4] Bolus, Willats, Rozell and Stanley (2021). *State-space optimal feedback control of optogenetically driven neural activity*. Journal of Neural Engineering 18, 036006. Additional source on observer-based implementation.

[5] Yin et al. (2022). *Cortical phase-amplitude coupling is key to the occurrence and treatment of freezing of gait*. Brain.

[6] K"ohler et al. (2024). *Dopamine and deep brain stimulation accelerate the neural dynamics of volitional action in Parkinson's disease*. Brain.

[7] Yin et al. (2024). *Generalized sleep decoding with basal ganglia signals in multiple movement disorders*. npj Digital Medicine.

[8] Liu et al. (2018). *A Compact Closed-Loop Optogenetics System Based on Artifact-Free Transparent Graphene Electrodes*. Frontiers in Neuroscience.

[9] Koeglsperger et al. (2026). *Adaptive Deep Brain Stimulation for Parkinson's Disease: Navigating the Roadblocks to Clinical Implementation*. Movement Disorders.

[10] Zaaimi et al. (2023). *Closed-loop optogenetic control of the dynamics of neural activity in non-human primates*. Nature Biomedical Engineering.

[11] Kathe et al. (2022). *Wireless closed-loop optogenetics across the entire dorsoventral spinal cord in mice*. Nature Biotechnology.

[12] Neumann et al. (2023). *Neurophysiological mechanisms of deep brain stimulation across spatiotemporal resolutions*. Brain.

[13] Rajamani et al. (2024). *Deep brain stimulation of symptom-specific networks in Parkinson's disease*. Nature Communications.

[14] De la Crompe et al. (2025). *Reevaluating the role of the STN-GPe network in generating beta synchronized oscillatory activity in parkinsonism*. Handbook of Parkinson's Disease Mechanisms, ch. 17.

[15] Nevado Holgado et al. (2010). *Conditions for the Generation of Beta Oscillations in the Subthalamic Nucleus-Globus Pallidus Network*. Journal of Neuroscience.

[16] Azizpour Lindi et al. (2024). *Synaptic Changes in Pallidostriatal Circuits Observed in the Parkinsonian Model Triggers Abnormal Beta Synchrony with Accurate Spatio-temporal Properties across the Basal Ganglia*. Journal of Neuroscience.

[17] Ortone et al. (2023). *Dopamine depletion leads to pathological synchronization of distinct basal ganglia loops in the beta band*. PLOS Computational Biology.